

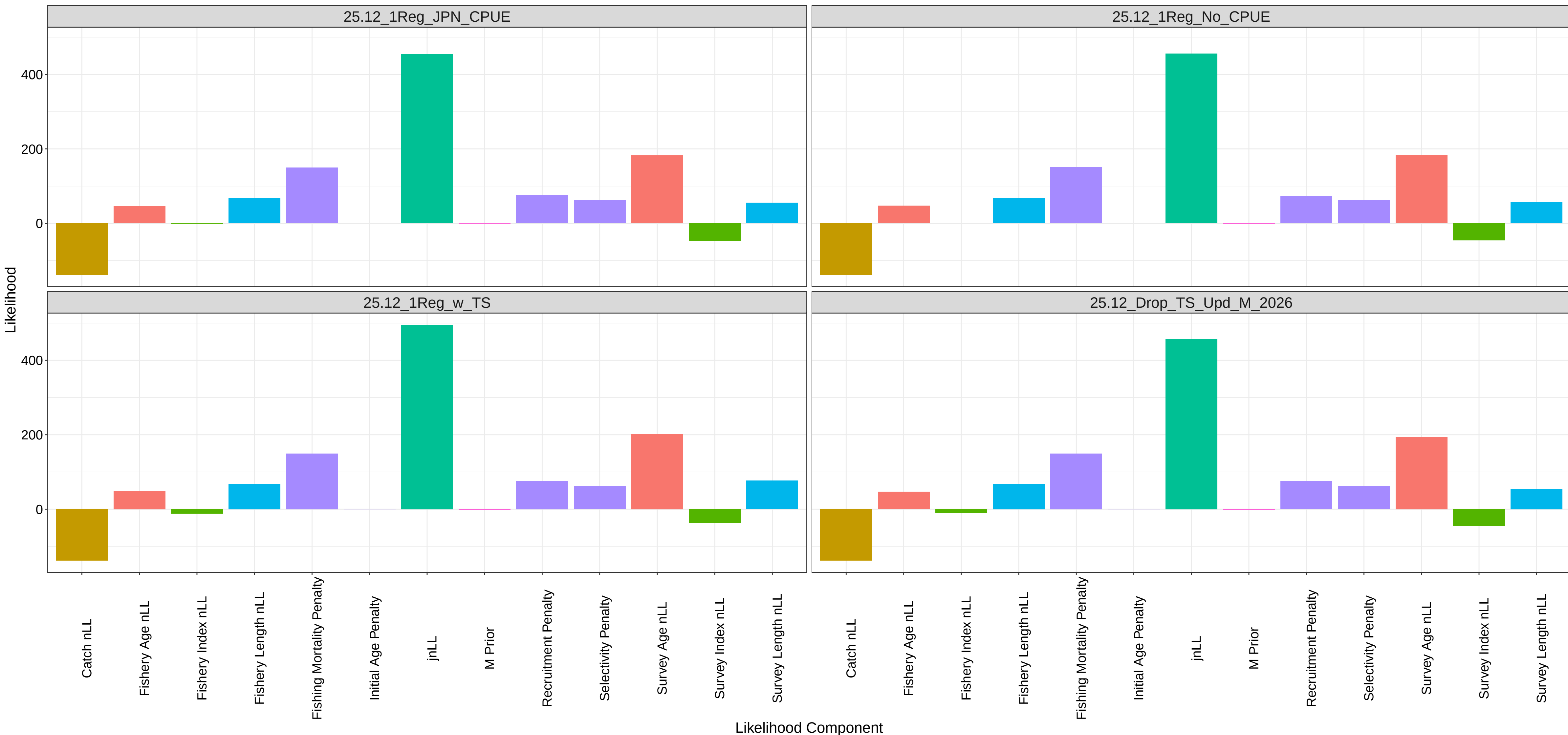
	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	25.12_Drop_TS_Upd_M_2026	1	206.6999	357.0208	0.04784668	39.55138	131.4549	0.07875	1.57240	0.57896	0.60758
2	25.12_1Reg_No_CPUE	1	221.0679	369.8268	0.04481018	41.69975	134.1211	0.07816	1.64827	0.59776	0.57329
3	25.12_1Reg_JPN_CPUE	1	222.3088	369.1939	0.04452147	42.31950	133.4034	0.07891	1.66644	0.60215	0.56421
4	25.12_1Reg_w_TS	1	189.0170	338.1359	0.05139846	37.78830	126.5573	0.07908	1.49353	0.55900	0.64996

	Model	jnLL	M Prior	Recruitment Penalty	Selectivity Penalty	Catch nLL	Fishing Mortality Penalty	Survey Index nLL	Fishery Index nLL	Initial Age Penalty	Survey Age nLL	Fishery Age nLL	Survey Length nLL	Fishery Length nLL
1	25.12_Drop_TS_Upd_M_2026	456.8247	-1.0189779	75.86092	62.56423	-138.3008	149.3833	-45.39642	-11.212232	0.2741517	194.6365	46.66608	55.19863	68.16928
2	25.12_1Reg_No_CPUE	455.7059	-0.9747952	73.15797	62.79468	-138.4294	150.4444	-45.63350	NA	0.1248466	183.4348	46.82344	55.77149	68.19207
3	25.12_1Reg_JPN_CPUE	454.4553	-0.8597552	75.92687	62.63891	-138.4152	149.3913	-46.24133	-0.759623	0.2660067	182.5290	46.61004	55.51924	67.84984
4	25.12_1Reg_w_TS	495.2505	-0.9806730	76.43750	62.51716	-138.2383	149.3609	-36.76800	-11.936829	0.2720866	202.2492	47.74680	76.56995	68.02074

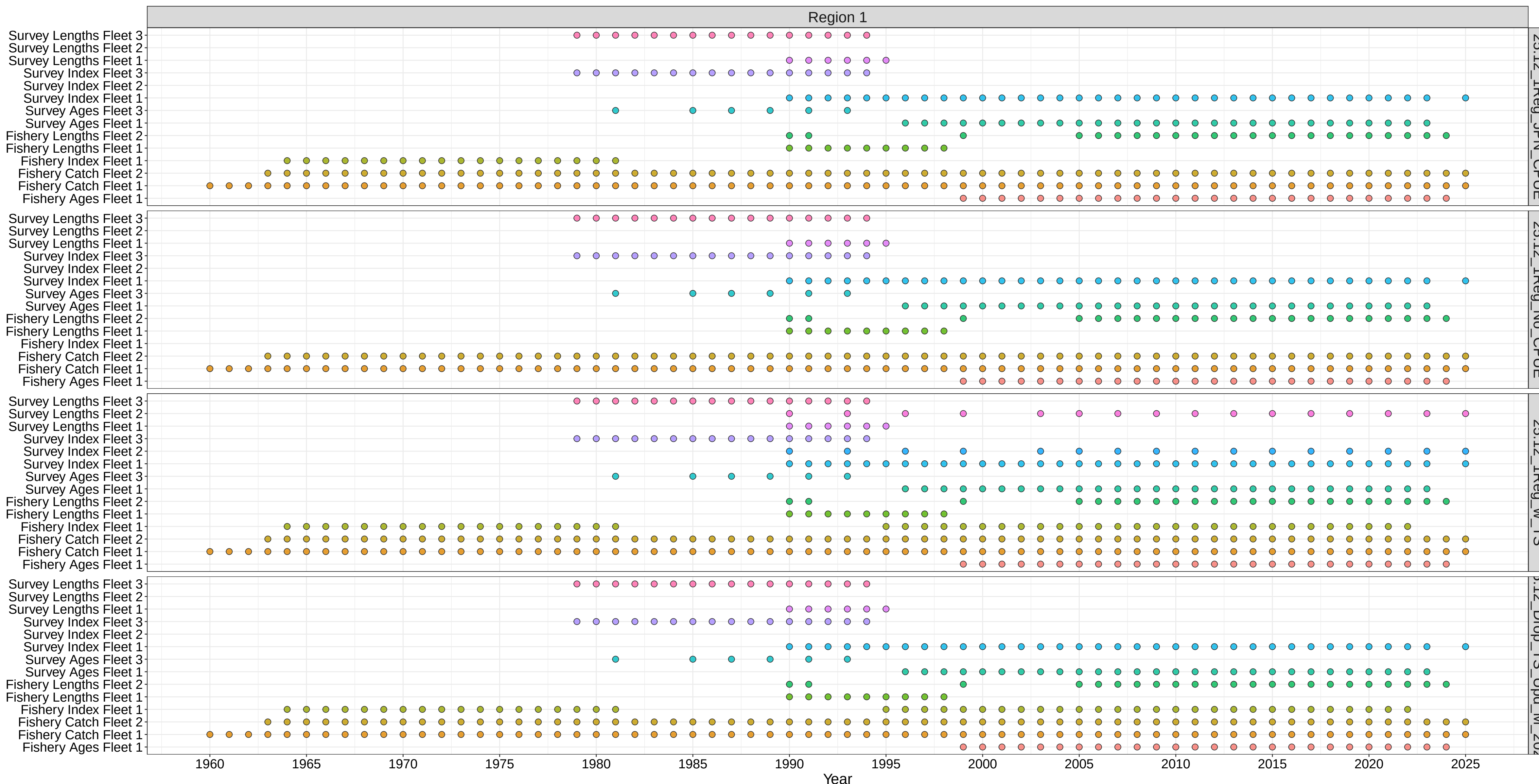
Negative Log-Likelihood

Type

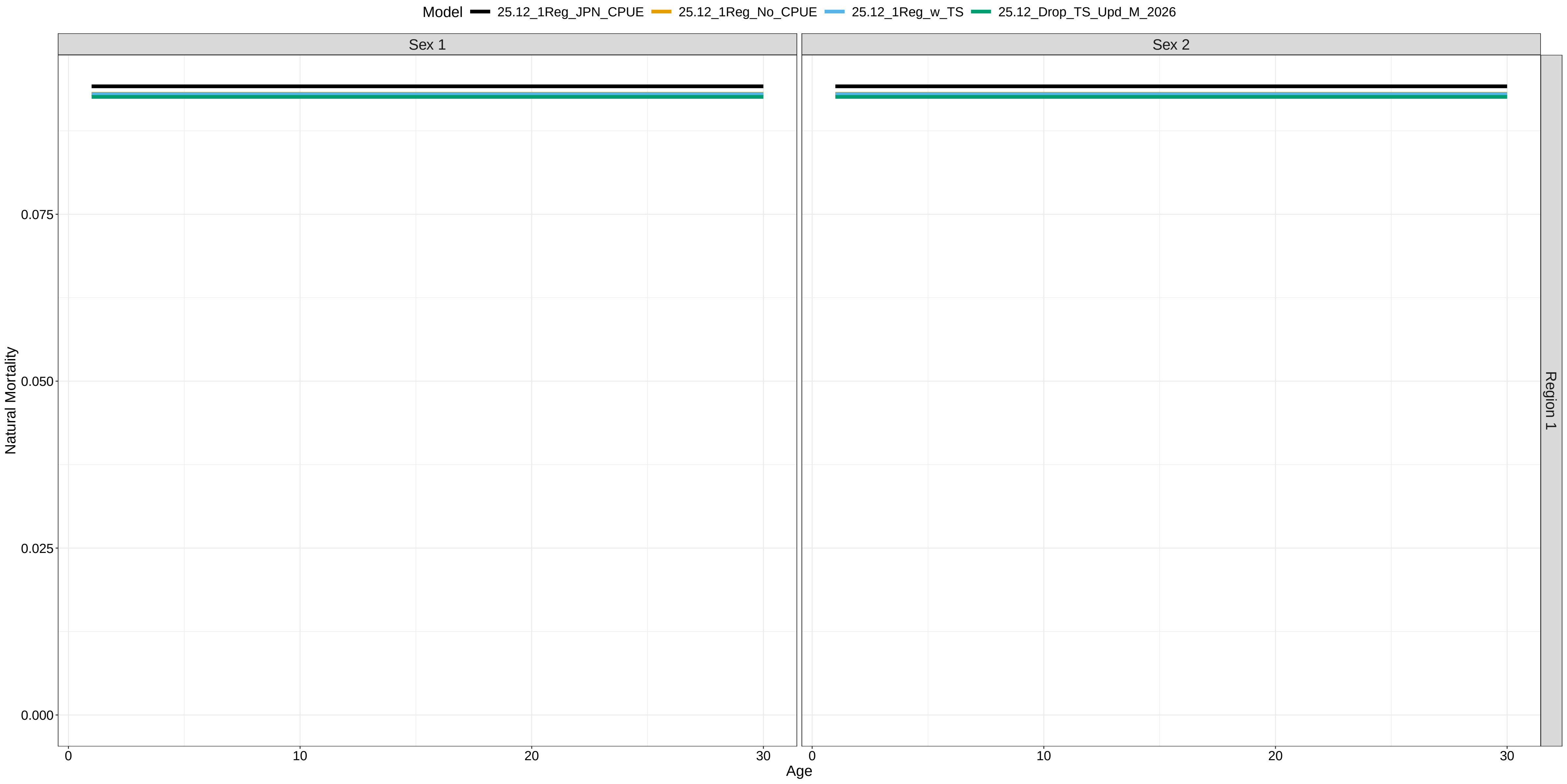
Age	Index	Length	Prior
Catch	jnLL	Penalty	



Data



Natural Mortality



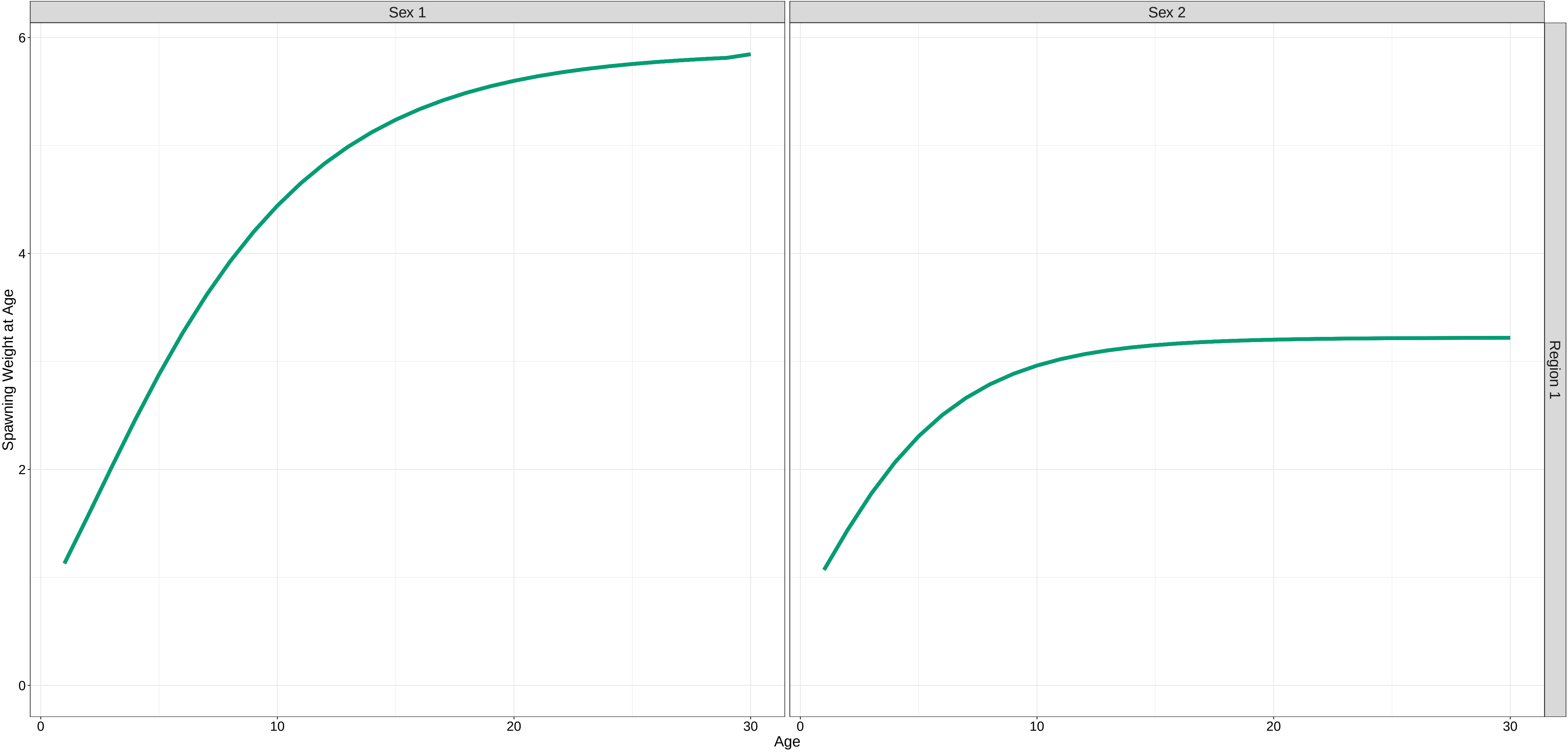
Weight-at-Age

Model 25.12_1Reg_JPN_CPUE 25.12_1Reg_No_CPUE 25.12_1Reg_w_TS 25.12_Drop_TS_Upd_M_2026

Sex 1

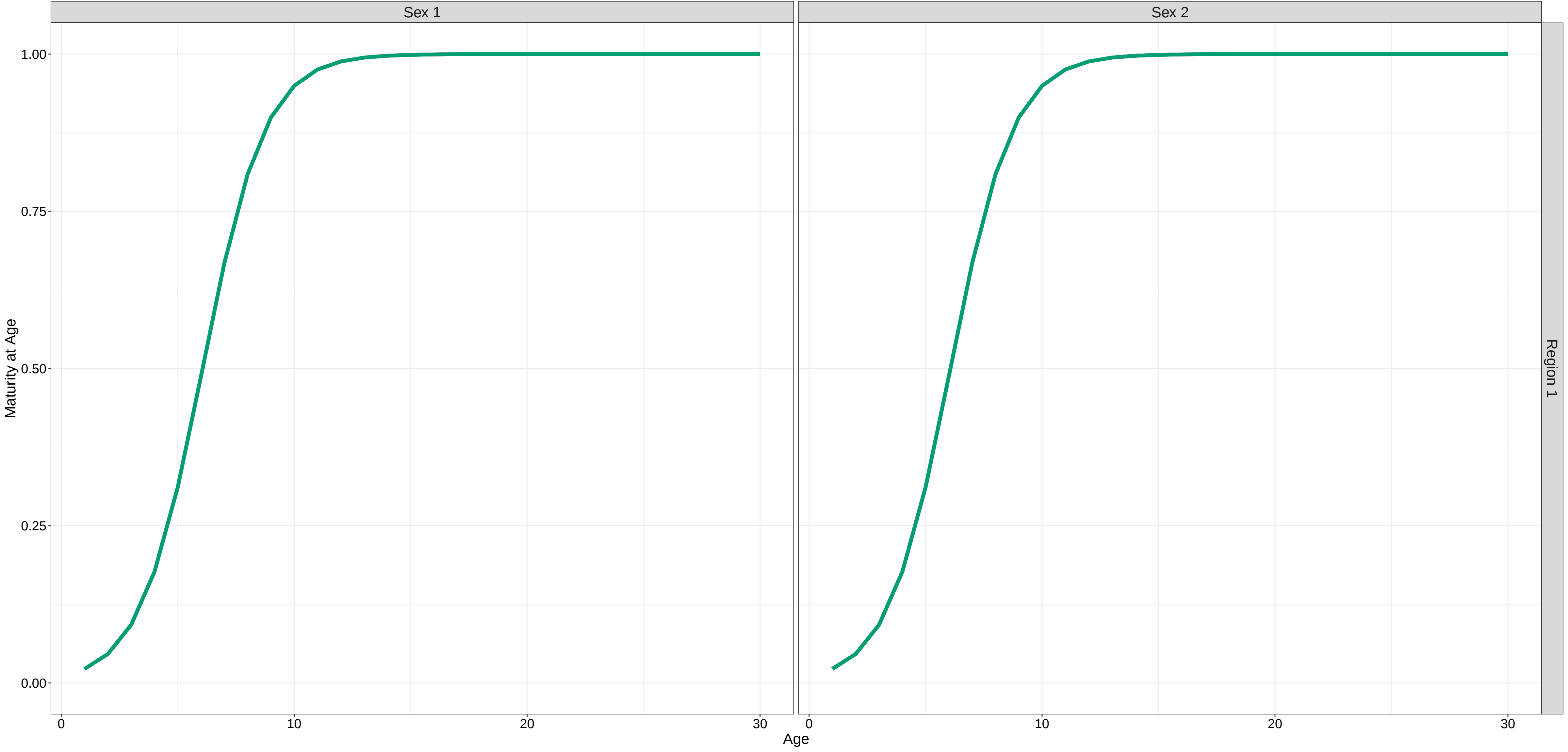
Sex 2

Region 1



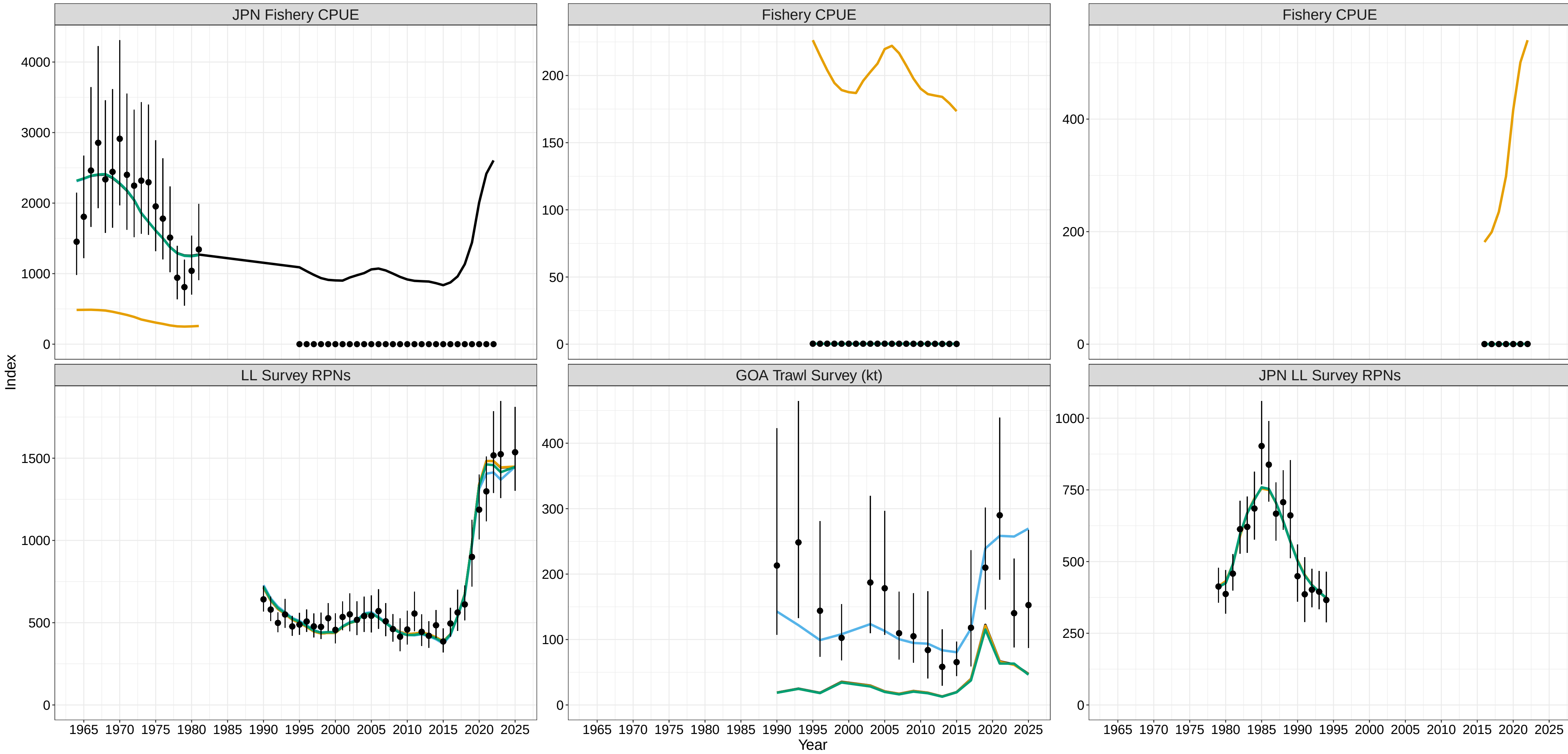
Maturity

Model 25.12_1Reg_JPN_CPUE 25.12_1Reg_No_CPUE 25.12_1Reg_w_TS 25.12_Drop_TS_Upd_M_2026



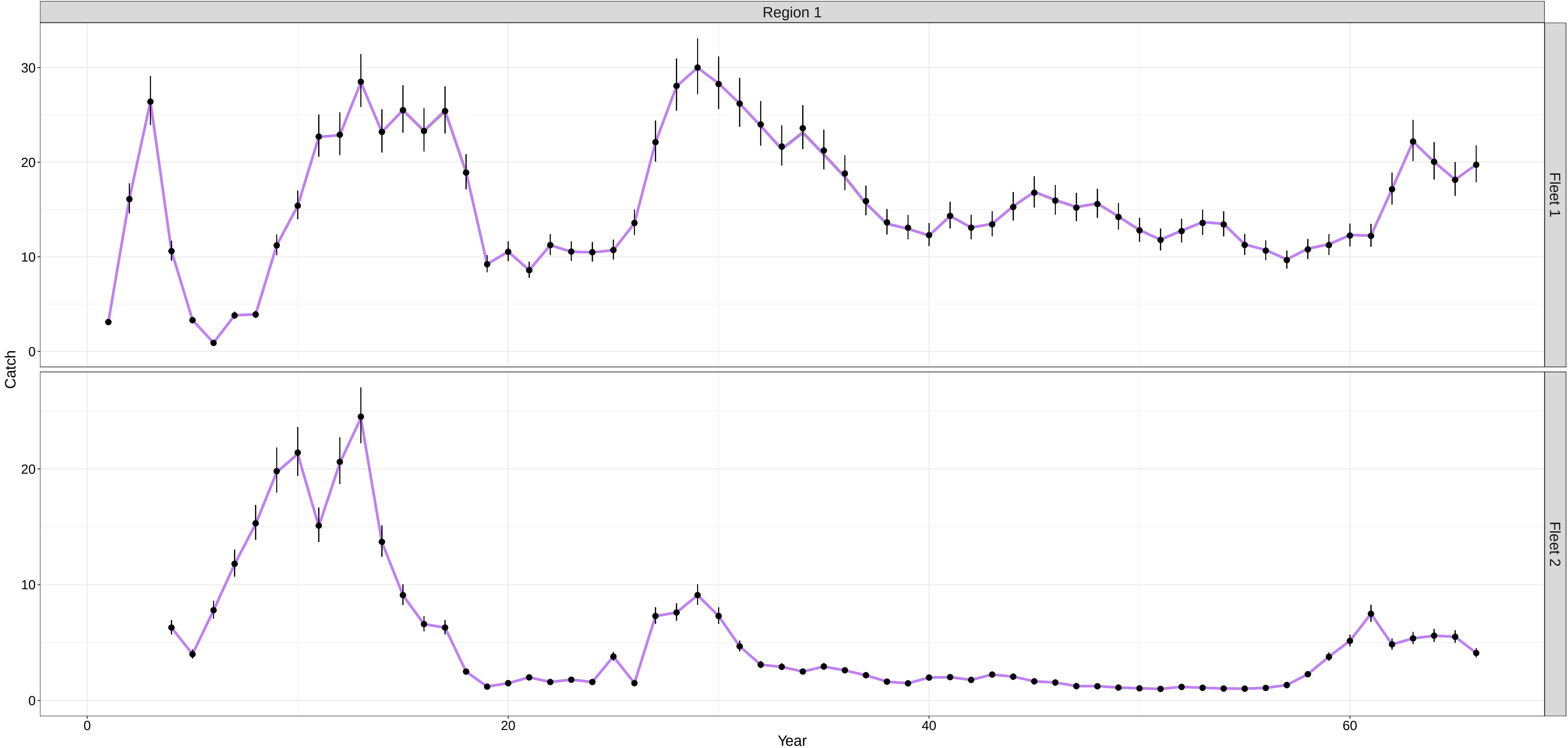
Index Fits

Model — 25.12_1Reg_JPN_CPUE — 25.12_1Reg_No_CPUE — 25.12_1Reg_w_TS — 25.12_Drop_TS_Upd_M_2026

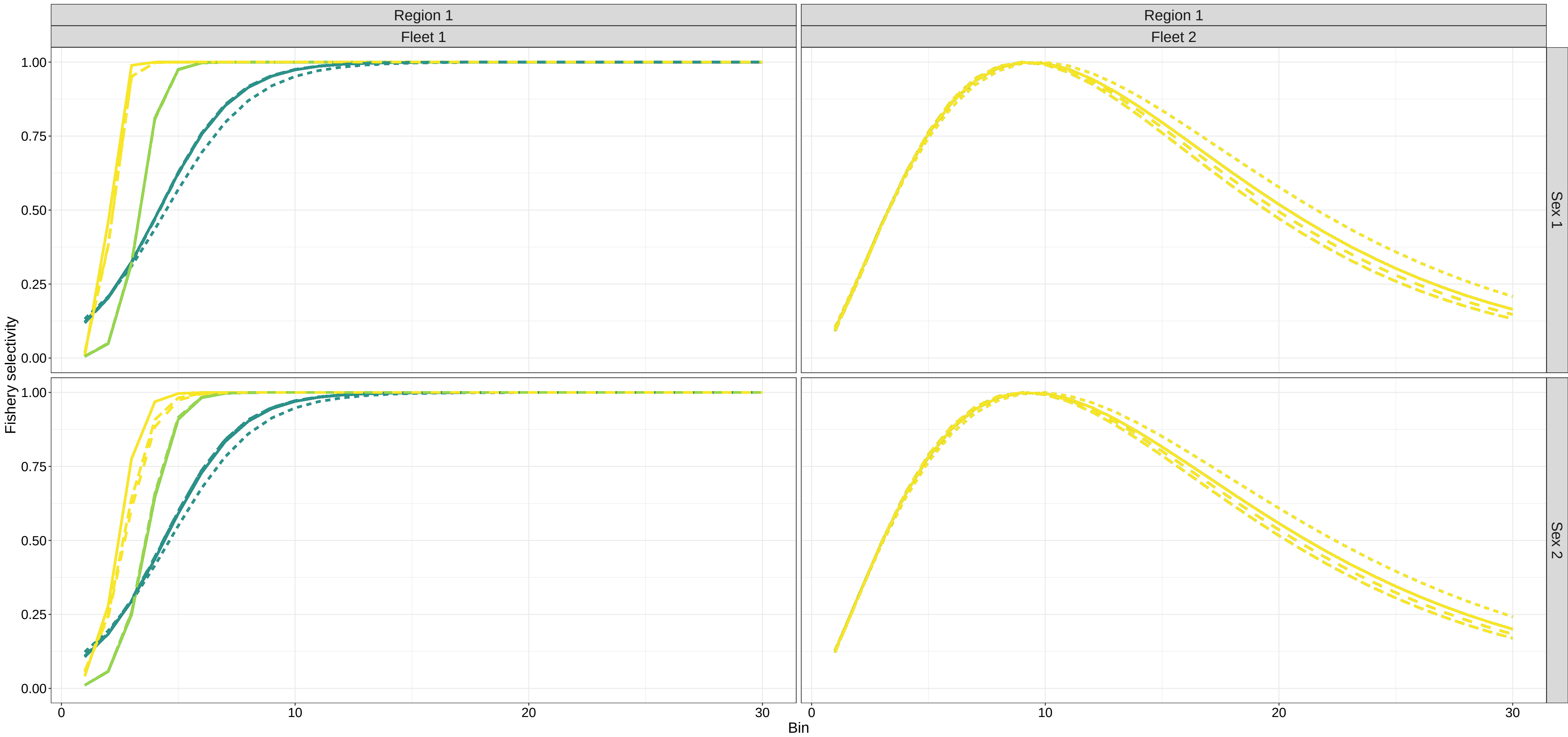
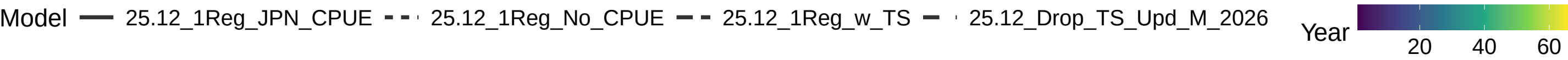


Catch

Model 25.12_1Reg_JPN_CPUE 25.12_1Reg_No_CPUE 25.12_1Reg_w_TS 25.12_Drop_TS_Upd_M_2026

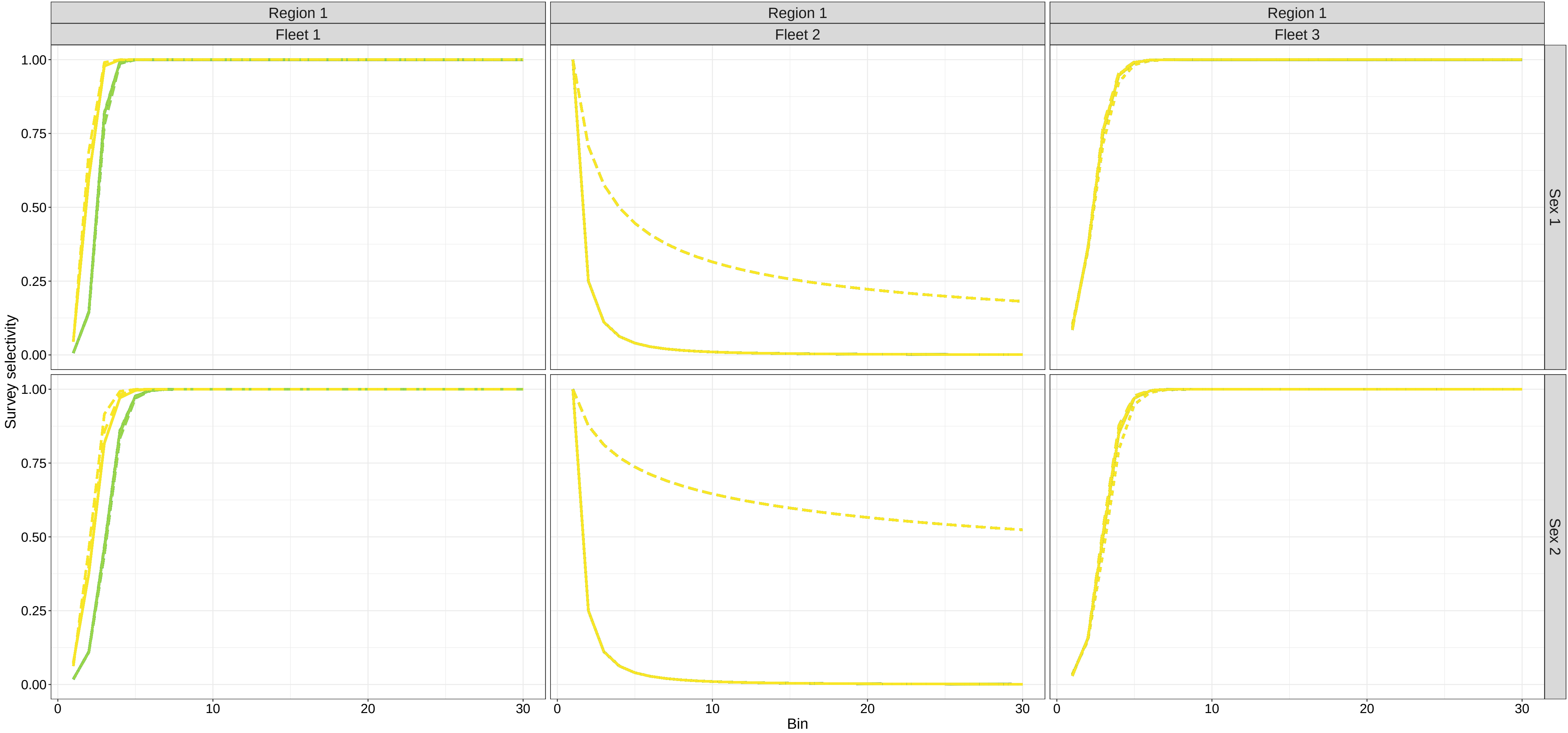


Fishery Selectivity



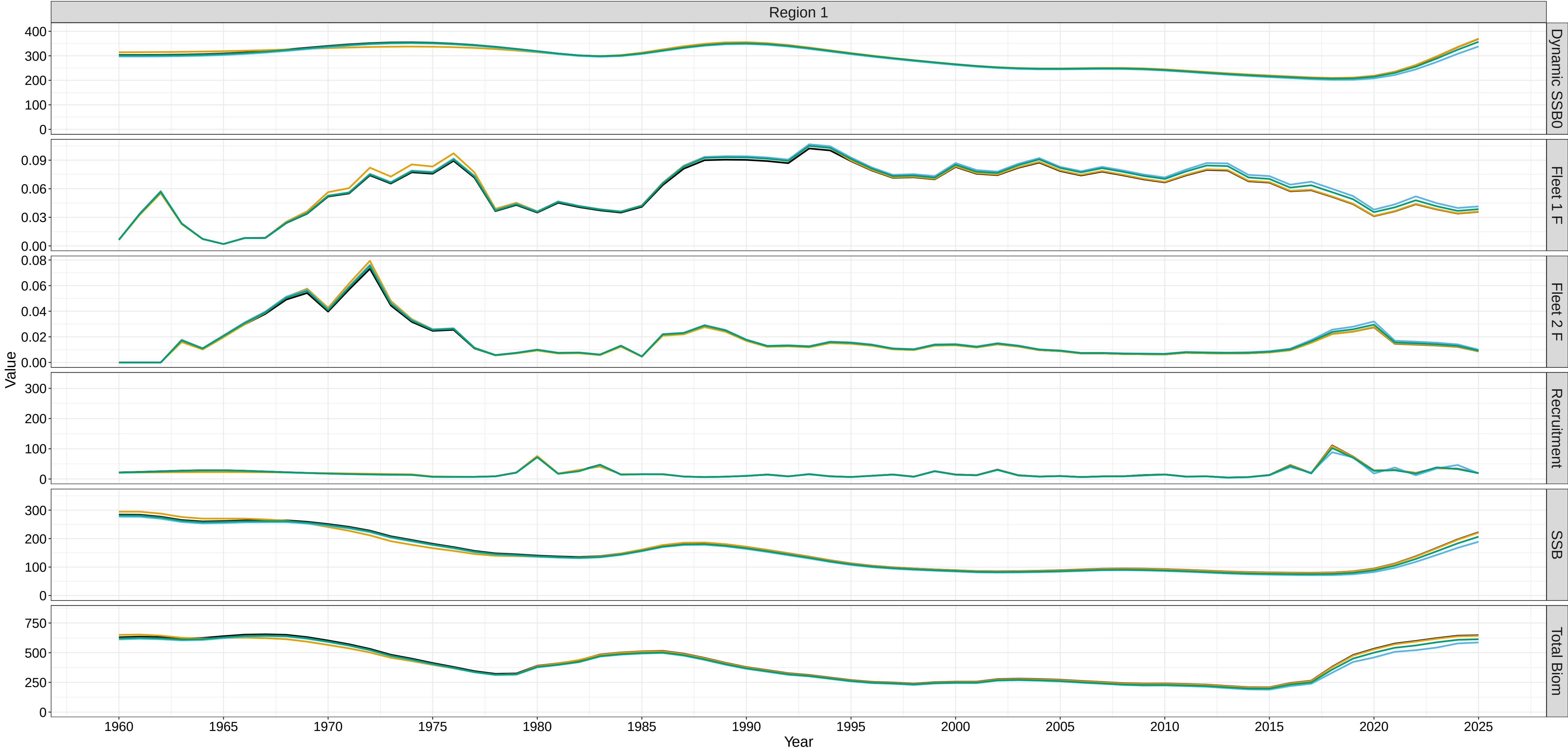
Survey Selectivity

Model — 25.12_1Reg_JPN_CPUE - - 25.12_1Reg_No_CPUE - - 25.12_1Reg_w_TS - · 25.12_Drop_TS_Upd_M_2026 Year 20 40 60



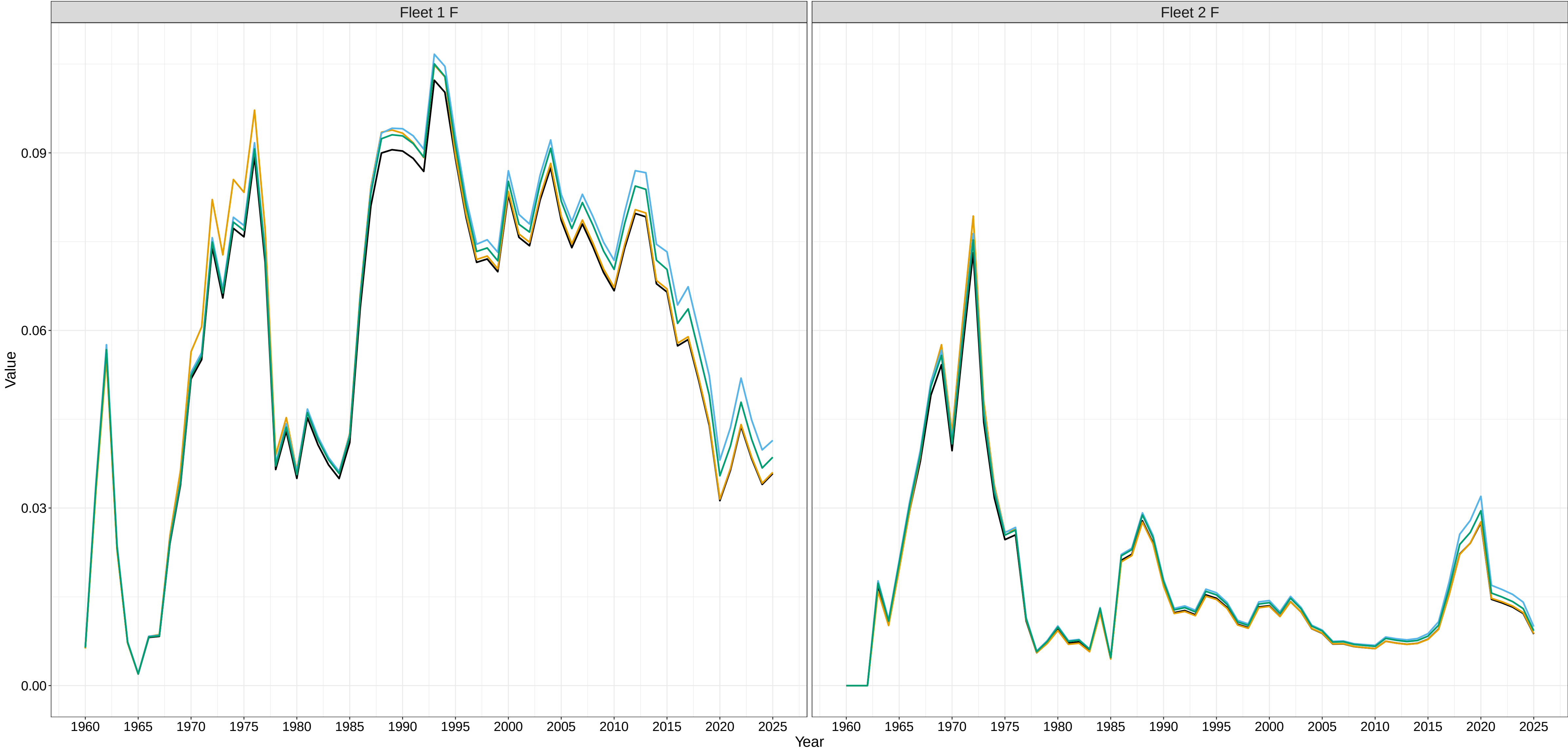
Time Series Plots

Model — 25.12_1Reg_JPN_CPUE — 25.12_1Reg_No_CPUE — 25.12_1Reg_w_TS — 25.12_Drop_TS_Upd_M_2026



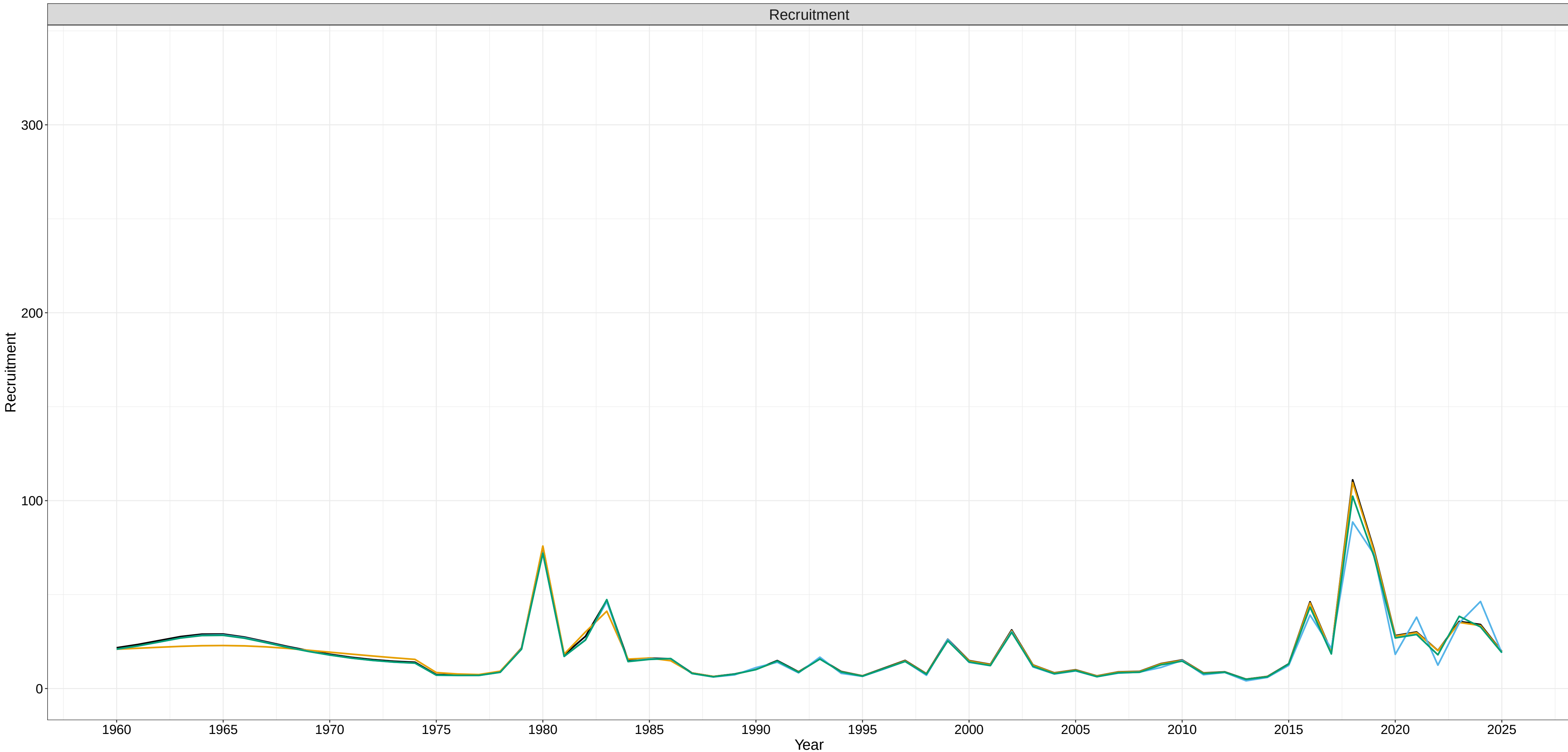
Fishing Mortality

Model — 25.12_1Reg_JPN_CPUE — 25.12_1Reg_No_CPUE — 25.12_1Reg_w_TS — 25.12_Drop_TS_Upd_M_2026



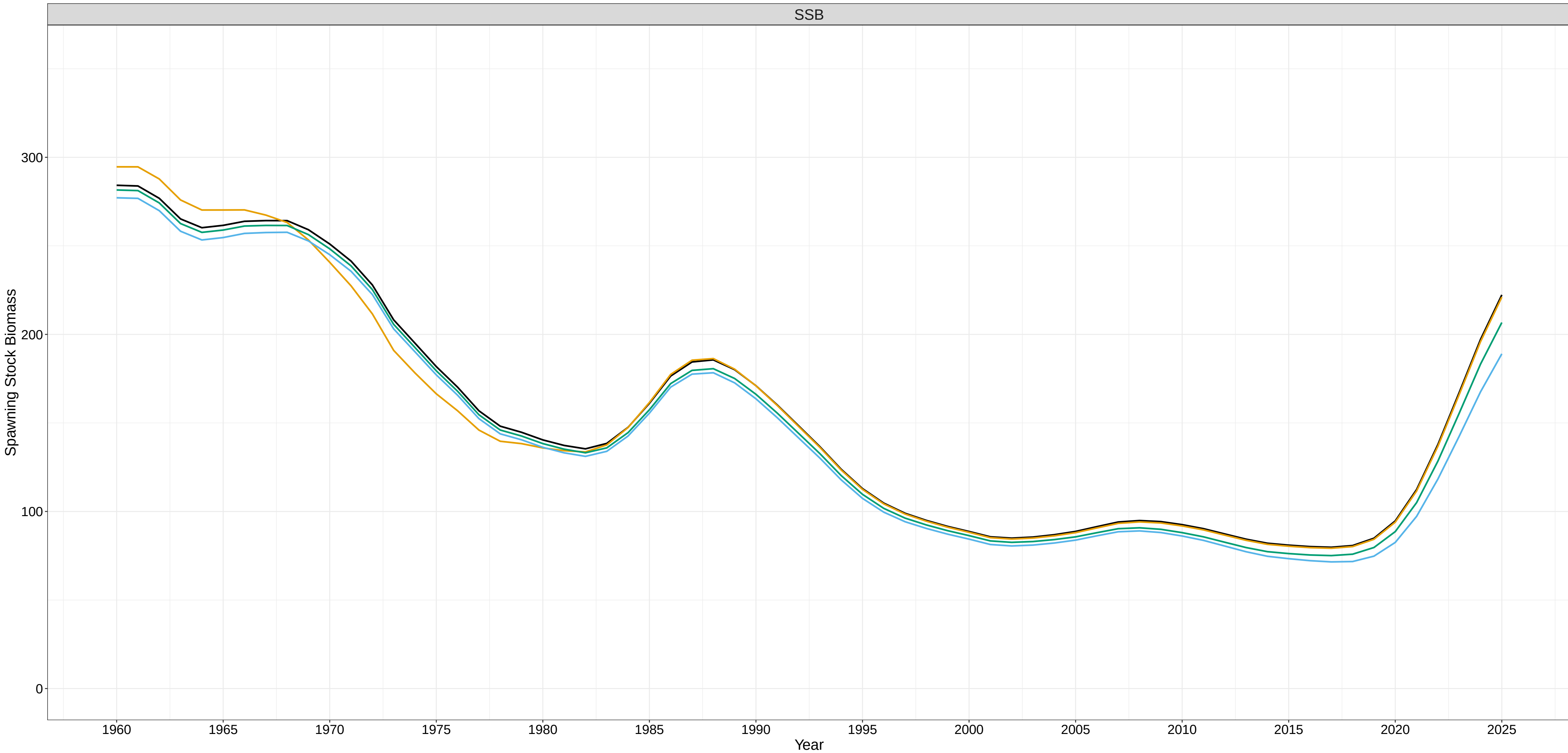
Recruitment

Model — 25.12_1Reg_JPN_CPUE — 25.12_1Reg_No_CPUE — 25.12_1Reg_w_TS — 25.12_Drop_TS_Upd_M_2026



SSB

Model — 25.12_1Reg_JPN_CPUE — 25.12_1Reg_No_CPUE — 25.12_1Reg_w_TS — 25.12_Drop_TS_Upd_M_2026



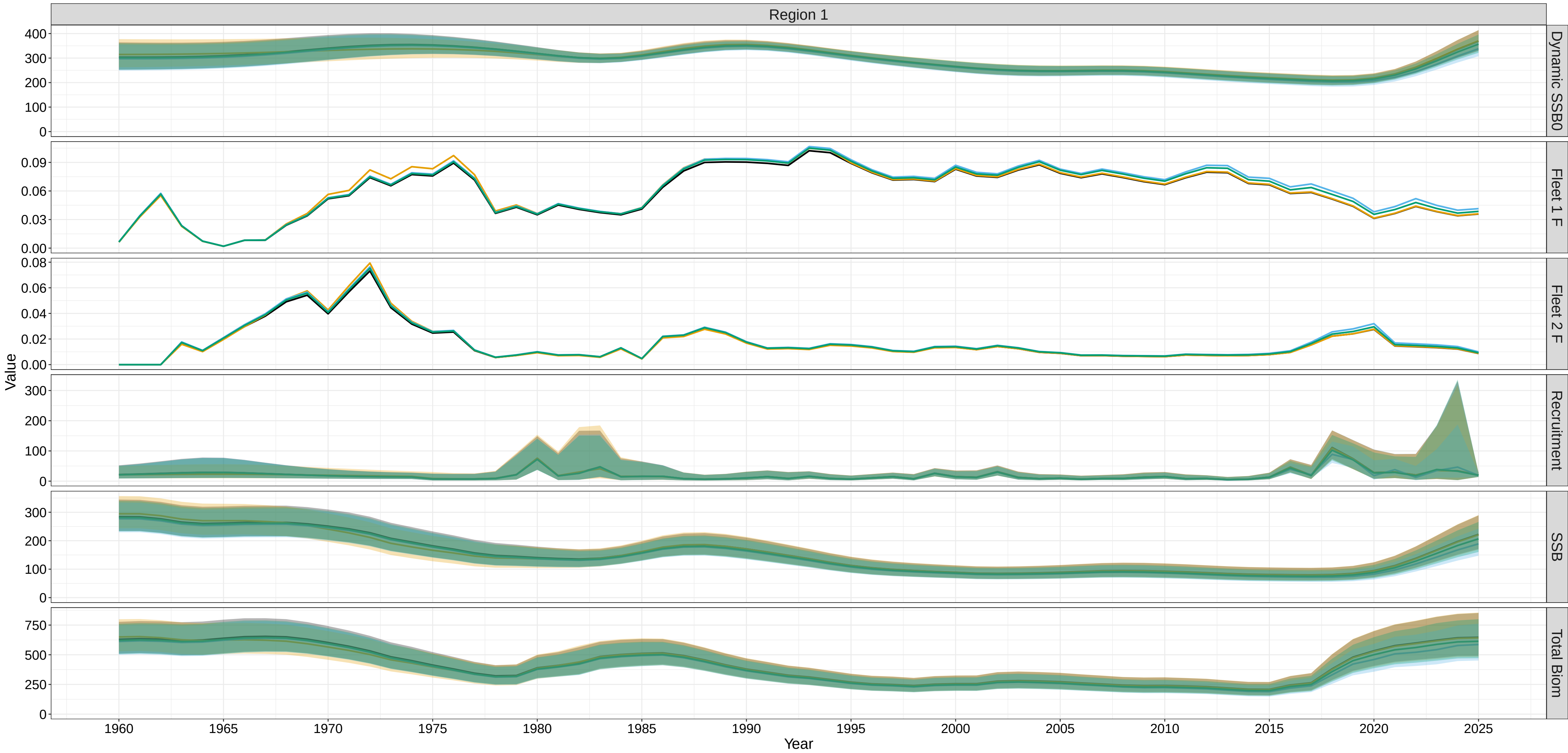
Biomass

Model — 25.12_1Reg_JPN_CPUE — 25.12_1Reg_No_CPUE — 25.12_1Reg_w_TS — 25.12_Drop_TS_Upd_M_2026



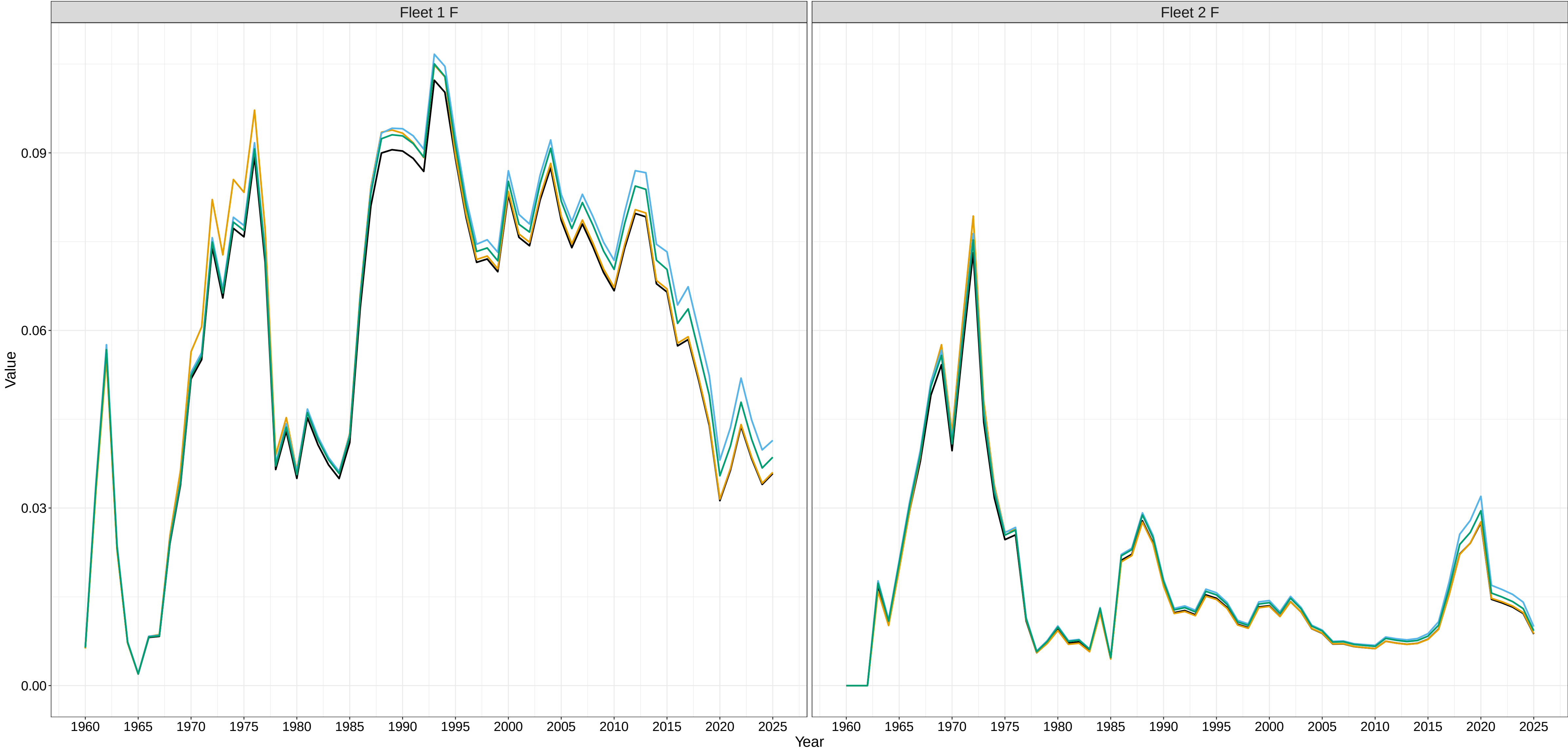
Time Series Plots

Model 25.12_1Reg_JPN_CPUE 25.12_1Reg_No_CPUE 25.12_1Reg_w_TS 25.12_Drop_TS_Upd_M_2026



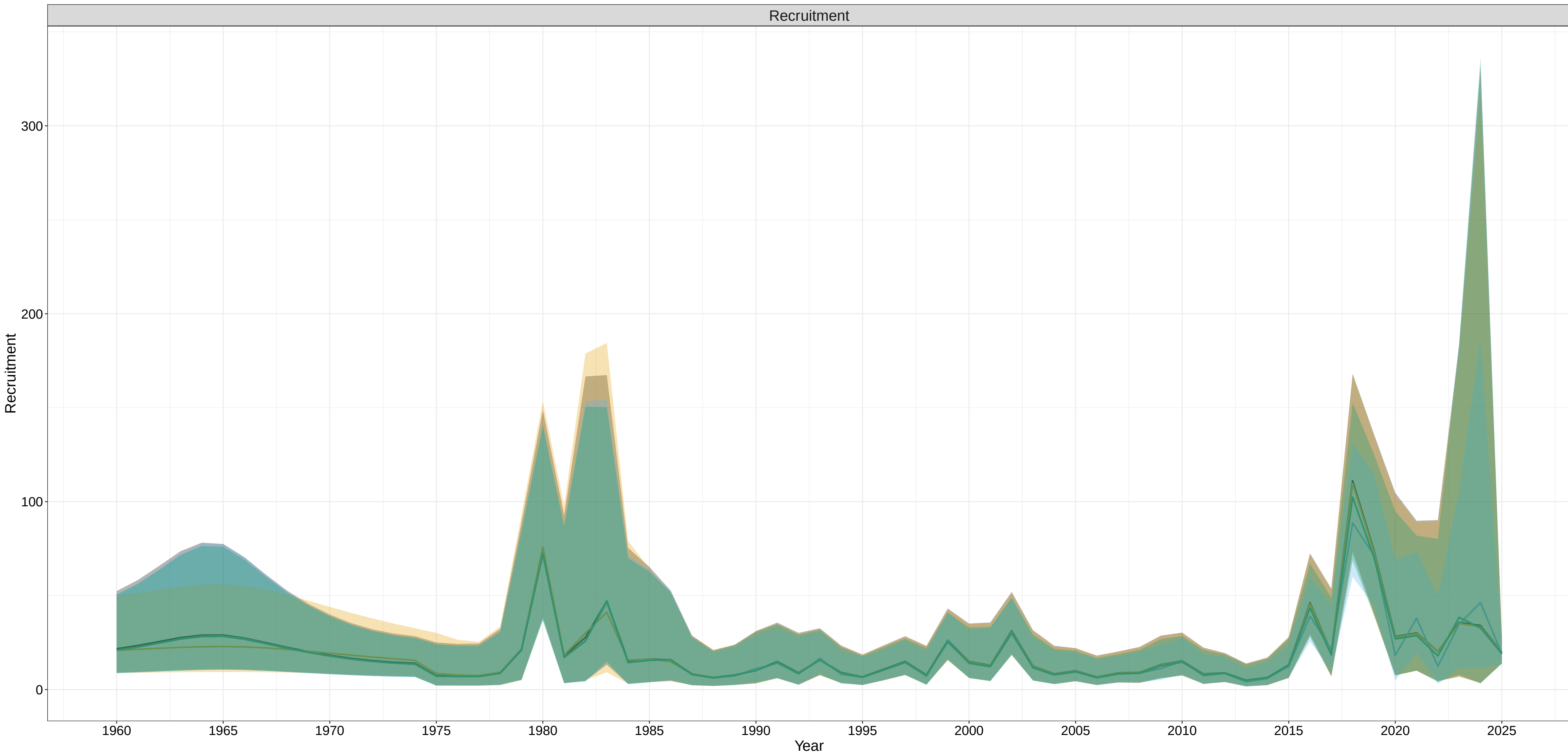
Fishing Mortality

Model — 25.12_1Reg_JPN_CPUE — 25.12_1Reg_No_CPUE — 25.12_1Reg_w_TS — 25.12_Drop_TS_Upd_M_2026



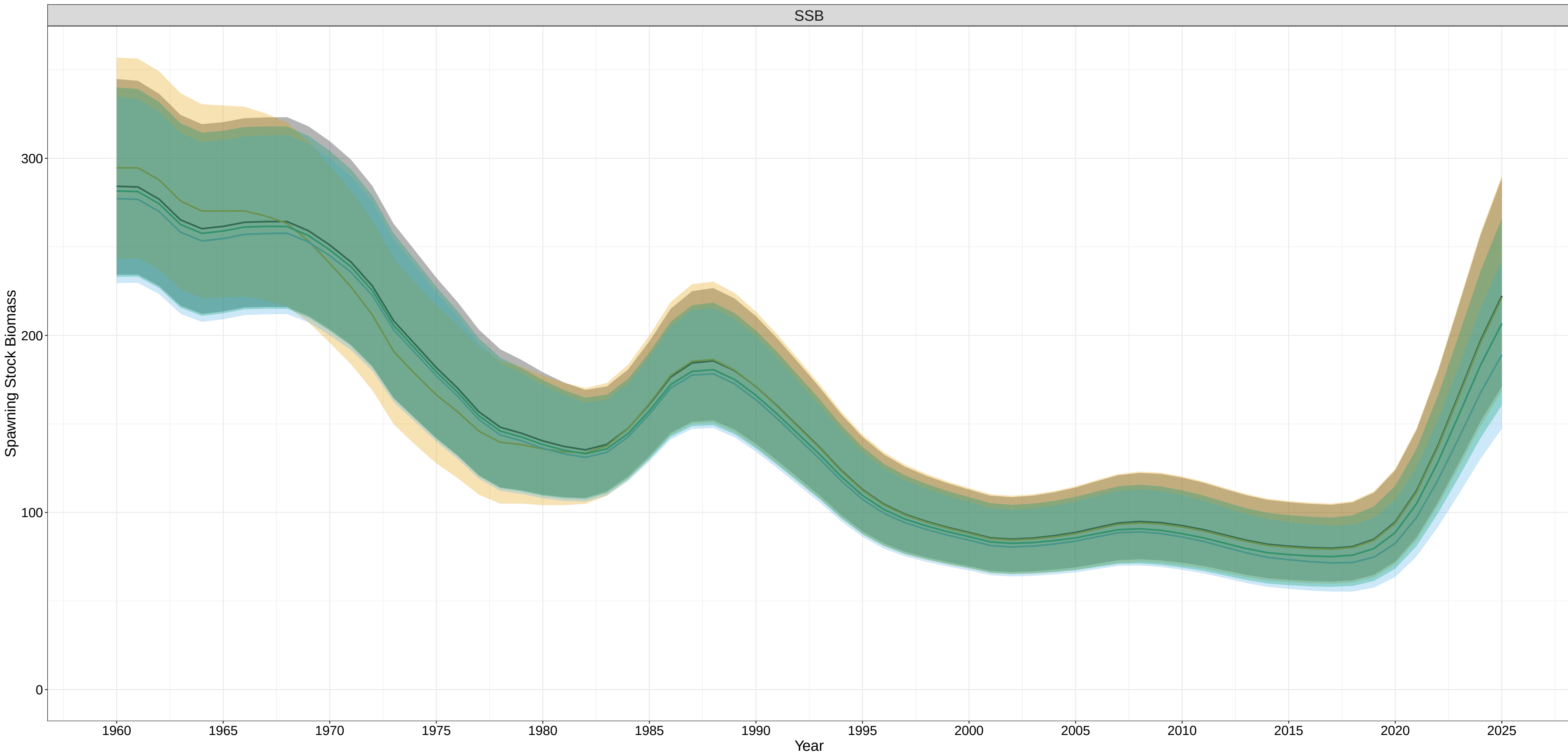
Recruitment

Model 25.12_1Reg_JPN_CPUE 25.12_1Reg_No_CPUE 25.12_1Reg_w_TS 25.12_Drop_TS_Upd_M_2026



SSB

Model 25.12_1Reg_JPN_CPUE 25.12_1Reg_No_CPUE 25.12_1Reg_w_TS 25.12_Drop_TS_Upd_M_2026



Biomass

Model 25.12_1Reg_JPN_CPUE 25.12_1Reg_No_CPUE 25.12_1Reg_w_TS 25.12_Drop_TS_Upd_M_2026

